WORKED SOLUTION

STEP II 1991, Question 14

Mechanics • Relative Velocity Pursuit

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Read the question before you go on. It is in the STEP Support Programme's question database:

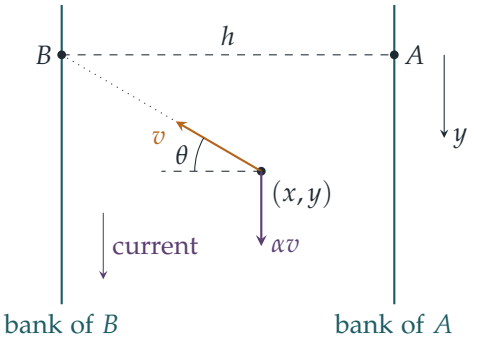
step.maths.org/questions/91-s2-q14

The key idea

The boat's ground velocity is its velocity through the water plus the velocity of the water. "Always pointing at B" is a geometric condition, $y = x\tan\theta$, and differentiating it gives a second expression for the same velocity. Matching the two produces a separable equation linking x and θ. The substitution $\tan\theta = \sinh(\alpha s)$ is chosen because it makes $\sec\theta + \tan\theta$ an exponential, which turns everything into simple exponentials of s.

Solution

We model the boat as a particle. Take B as the origin, with x measured from B's bank towards A and y measured downstream, so that A is at $(h, 0)$ and the boat starts there.



The velocity in terms of θ

The boat points towards B's bank, at angle θ to the direction AB, angled upstream (to counter the drift). Its velocity relative to the water is therefore v along the unit vector $-(\cos\theta, \sin\theta)$ in our (x, y) components. Adding the current, which is αv in the

y-direction,

$$\begin{pmatrix}\dot{x}\\ \dot{y}\end{pmatrix} = \begin{pmatrix}-v\cos\theta\\ \alpha v - v\sin\theta\end{pmatrix}.$$

Pointing at B

The direction from the boat at (x, y) to B is along $(-x, -y)$, and this makes angle θ with AB exactly when $\tan\theta = y/x$. So, throughout the motion,

$$y = x\tan\theta.$$

Differentiating with respect to t by the product and chain rules,

$$\dot{y} = \tan\theta\,\frac{\mathrm{d}x}{\mathrm{d}t} + x\sec^2\theta\,\frac{\mathrm{d}\theta}{\mathrm{d}t},$$

so the velocity vector $(\dot{x}, \dot{y})$ has the required form.

The differential equation for x and θ

Now we match the two expressions. The x-components give $\dot{x} = -v\cos\theta$. Putting this into the y-component,

$$\begin{aligned}\tan\theta\,(-v\cos\theta) + x\sec^2\theta\,\dot{\theta} &= \alpha v - v\sin\theta\\ x\sec^2\theta\,\dot{\theta} &= \alpha v, \qquad (*)\end{aligned}$$

since $\tan\theta\cos\theta = \sin\theta$ and the $-v\sin\theta$ terms cancel. In particular $\dot{\theta} > 0$, so θ increases steadily from 0 and we may treat x as a function of θ. Dividing,

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \frac{\dot{x}}{\dot{\theta}} = \frac{-v\cos\theta\cdot x\sec^2\theta}{\alpha v} = -\frac{x\sec\theta}{\alpha}, \qquad \text{so} \qquad \alpha\frac{\mathrm{d}x}{\mathrm{d}\theta} = -x\sec\theta,$$

as required.

Solving for x

The equation separates (we have $x > 0$ until the boat arrives):

$$\alpha\int\frac{\mathrm{d}x}{x} = -\int\sec\theta\,\mathrm{d}\theta \quad\Longrightarrow\quad \alpha\ln x = -\ln(\sec\theta + \tan\theta) + C,$$

using the standard integral $\int\sec\theta\,\mathrm{d}\theta = \ln(\sec\theta + \tan\theta)$, valid for $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$. At the start the boat is at A, where $x = h$ and $\theta = 0$, so $C = \alpha\ln h$. Hence

$$\alpha\ln\frac{x}{h} = -\ln(\sec\theta + \tan\theta), \qquad \text{that is,} \qquad \left(\frac{x}{h}\right)^{\alpha} = (\sec\theta + \tan\theta)^{-1}.$$

The substitution $\tan\theta = \sinh(\alpha s)$

Ask yourself What is $\sec\theta$ in terms of s?

Since $\sec^2\theta = 1 + \tan^2\theta = 1 + \sinh^2(\alpha s) = \cosh^2(\alpha s)$, and both $\sec\theta$ and $\cosh(\alpha s)$ are positive (as $0 \le \theta < \frac{\pi}{2}$), we get $\sec\theta = \cosh(\alpha s)$. Then

$$\sec\theta + \tan\theta = \cosh(\alpha s) + \sinh(\alpha s) = \mathrm{e}^{\alpha s},$$

so $(x/h)^\alpha = \mathrm{e}^{-\alpha s}$, and taking αth roots, $x = h\mathrm{e}^{-s}$.

For the rate equation, differentiate $\tan\theta = \sinh(\alpha s)$ with respect to t:

$$\sec^2\theta\,\dot\theta = \alpha\cosh(\alpha s)\,\dot s.$$

Substituting this and $x = h\mathrm{e}^{-s}$ into $(*)$ gives $h\mathrm{e}^{-s}\cdot\alpha\cosh(\alpha s)\,\dot s = \alpha v$, and dividing by α,

$$h\mathrm{e}^{-s}\cosh(\alpha s)\frac{\mathrm{d}s}{\mathrm{d}t} = v,$$

as required.

The time of crossing

At A, $\theta = 0$ so $s = 0$. Since $\dot s > 0$, s increases, and the boat reaches B when $x = h\mathrm{e}^{-s} \to 0$, that is, as $s \to \infty$. (It really does arrive at B: $y = x\tan\theta = h\mathrm{e}^{-s}\sinh(\alpha s) \to 0$ because $\alpha < 1$.) Separating the variables, the crossing time is

$$\begin{aligned} T &= \frac{h}{v}\int_0^\infty \mathrm{e}^{-s}\cosh(\alpha s)\,\mathrm{d}s = \frac{h}{2v}\int_0^\infty \left(\mathrm{e}^{-(1-\alpha)s} + \mathrm{e}^{-(1+\alpha)s}\right)\mathrm{d}s \\ &= \frac{h}{2v}\left(\frac{1}{1-\alpha} + \frac{1}{1+\alpha}\right) = \frac{h}{2v}\cdot\frac{2}{1-\alpha^2} = \frac{h}{v(1-\alpha^2)}, \end{aligned}$$

where both integrals converge because $1 - \alpha > 0$. So the time of crossing is $hv^{-1}(1-\alpha^2)^{-1}$.

Checks

- With no current ($\alpha = 0$) the boat heads straight across and $T = h/v$, as it should. As $\alpha \to 1$ the time blows up: a boat no faster than the current can never make headway against it near B, where it is heading straight upstream.
- We stepped the motion numerically (fourth-order Runge–Kutta, steering at B at every step). With $h = v = 1$ and $\alpha = \frac{1}{2}$ the boat arrives at $t \approx 1.33333$, matching $1/(1-\frac{1}{4}) = \frac{4}{3}$. Along the path, $(x/h)^\alpha(\sec\theta + \tan\theta)$ stayed equal to 1 to about 10^{-13}.

Common slips

- Getting the sign of the heading's y-component wrong. The boat drifts downstream, so pointing at B means pointing partly upstream; with $\theta > 0$ that component is $-v\sin\theta$.

- Forgetting to justify $\sec\theta = +\cosh(\alpha s)$: it needs θ to stay below $\frac{\pi}{2}$, which holds because θ is the angle of the line from B to a point with $x > 0$.

Beyond STEP

This is a classical *pursuit curve* problem: one object steers continually towards a target, and the path is found from a differential equation for the direction of motion. The same ideas describe a dog chasing a moving owner, or a missile homing on a target, and they lead naturally to curves such as the tractrix. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to build fluency with relative velocity and setting up equations of motion in moving frames.

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